Let’s go through the answers

Hand in your paper first, then follow along

Which question can whole-genome sequencing answer that standard culture + AST cannot?

a) Which antibiotic should this patient receive today

b) Whether isolates from several patients are the same strain ✓

c) Whether the organism is Gram-positive

d) Whether the patient has a fever

WHY

Culture and AST guide treatment for this patient. Only the genome shows whether isolates are the **same strain**.

Do you need to own a sequencer to do genomic surveillance? (Yes / No) Briefly explain why.

ANSWER

No. Isolates or DNA can be sent to a partner. The local skills are choosing isolates, recording metadata, analysing and acting on results.

Name two bacterial species that commonly cause surgical site infections.

ANSWER

Any two of: *S. aureus*, *E. coli*, *K. pneumoniae*, *P. aeruginosa*, *A. baumannii*, *Enterobacter*, *Enterococcus*.

An Illumina sequencing run typically produces reads that are:

a) 100–300 bases long and highly accurate ✓

b) 10,000+ bases long

c) the whole genome in one read

d) protein sequences

WHY

Illumina gives short, very accurate reads. Long reads come from Nanopore.

Which technology is better at showing whether a resistance gene is on a plasmid?

a) Illumina short reads alone

b) Oxford Nanopore long reads ✓

c) AST

d) MLST

WHY

Long reads can span a whole plasmid, so you can see which genes sit on it.

A genome is 5 million bases. You have 1 million reads of 150 bases. What is the approximate coverage?

a) 3×

b) 30× ✓

c) 150×

d) 300×

WHY

1,000,000 × 150 ÷ 5,000,000 = **30×**

What does a FASTQ file contain that a FASTA file does not?

a) Sequence names

b) DNA letters

c) Quality scores ✓

d) Gene names

WHY

FASTQ = sequence + a **quality** line for every read. FASTA = sequence only.

After assembly, which result usually means a *better* assembly?

a) More contigs and a smaller N50

b) Fewer contigs and a larger N50 ✓

c) A total length twice the expected genome size

d) No contigs

WHY

Fewer, longer pieces. A total length twice the expected size suggests contamination.

The gene *mecA* in *S. aureus* indicates:

a) Vancomycin resistance

b) MRSA (methicillin resistance) ✓

c) Carbapenem resistance

d) A virulence toxin

WHY

mecA makes an altered target (PBP2a), so almost all β-lactams stop working.

`blaNDM-1` is an example of:

a) An ESBL

b) A carbapenemase ✓

c) An efflux pump

d) A housekeeping gene

WHY

NDM = New Delhi metallo-β-lactamase: it destroys carbapenems.

The genome shows no known resistance gene, but AST says “resistant”. Give one possible explanation.

ANSWER

Any of: an unknown mechanism · a mutation not in the database · low coverage missed the gene · mixed culture · lab error.

Two *Klebsiella* isolates are ST15 and ST307. Could they be part of the same recent transmission chain?

a) Yes, very likely

b) No, they are different strain types ✓

c) Only if they are both ESBL

WHY

Different STs are not one recent transmission chain. (They could still share a plasmid.)

On a phylogenetic tree, what tells you how genetically different two isolates are?

a) Their order from top to bottom

b) The branch lengths between them ✓

c) The colour of the labels

WHY

Branch length = genetic change. Top-to-bottom order means nothing: branches can rotate.

Four isolates are 2–6 SNPs apart and the patients overlapped on the same ward. This most strongly suggests:

a) Unrelated infections

b) Possible transmission / outbreak ✓

c) Contamination of the sequencer

WHY

Very few SNPs **and** an epidemiological link: possible transmission.

List three pieces of metadata that must be recorded with every isolate for surveillance.

ANSWER

Any three of: collection date · facility · ward · specimen type · surgery type and date · AST results · organism · coded patient ID.